CRYSS

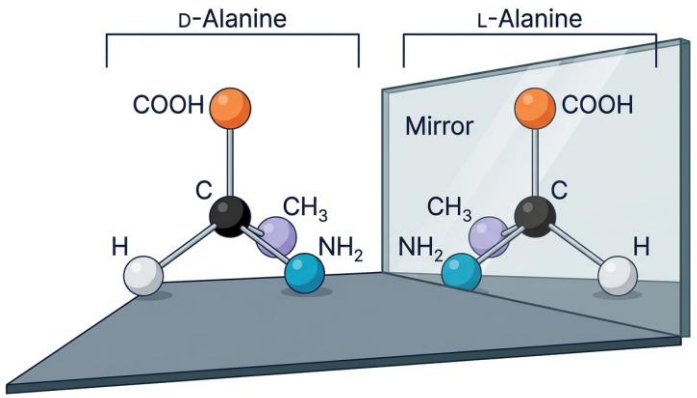


CRYstallization Screening System

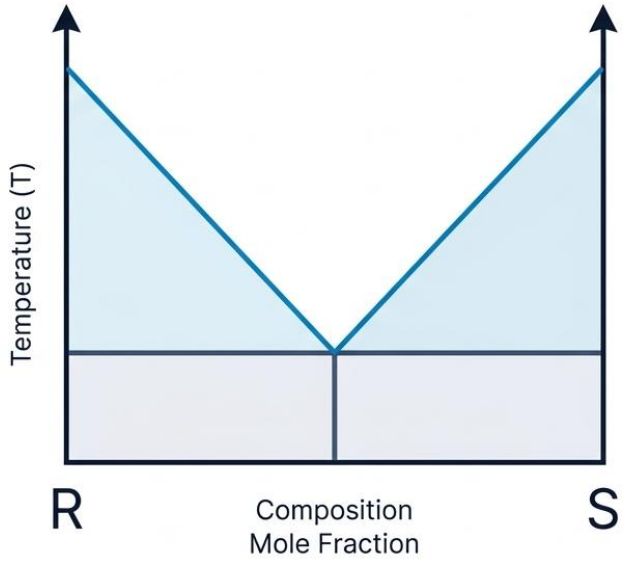
www.cryss.co.uk

12-Oct-2026 (Updated 9-Nov-2026)

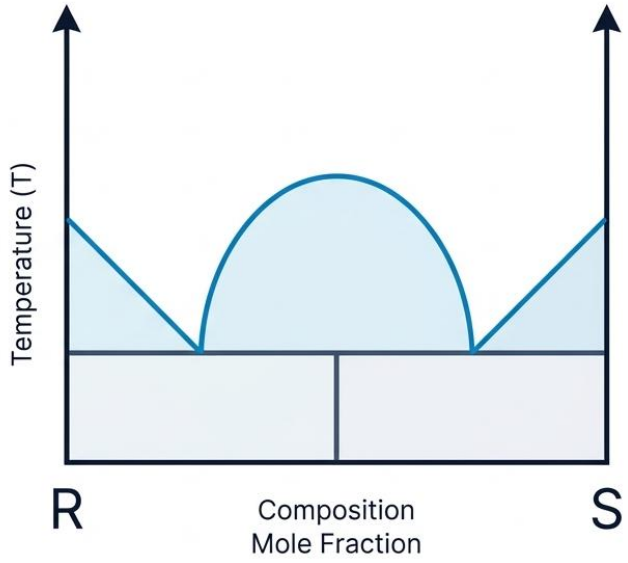
Types of Racemate



Melting Point Phase Diagram



Conglomerate

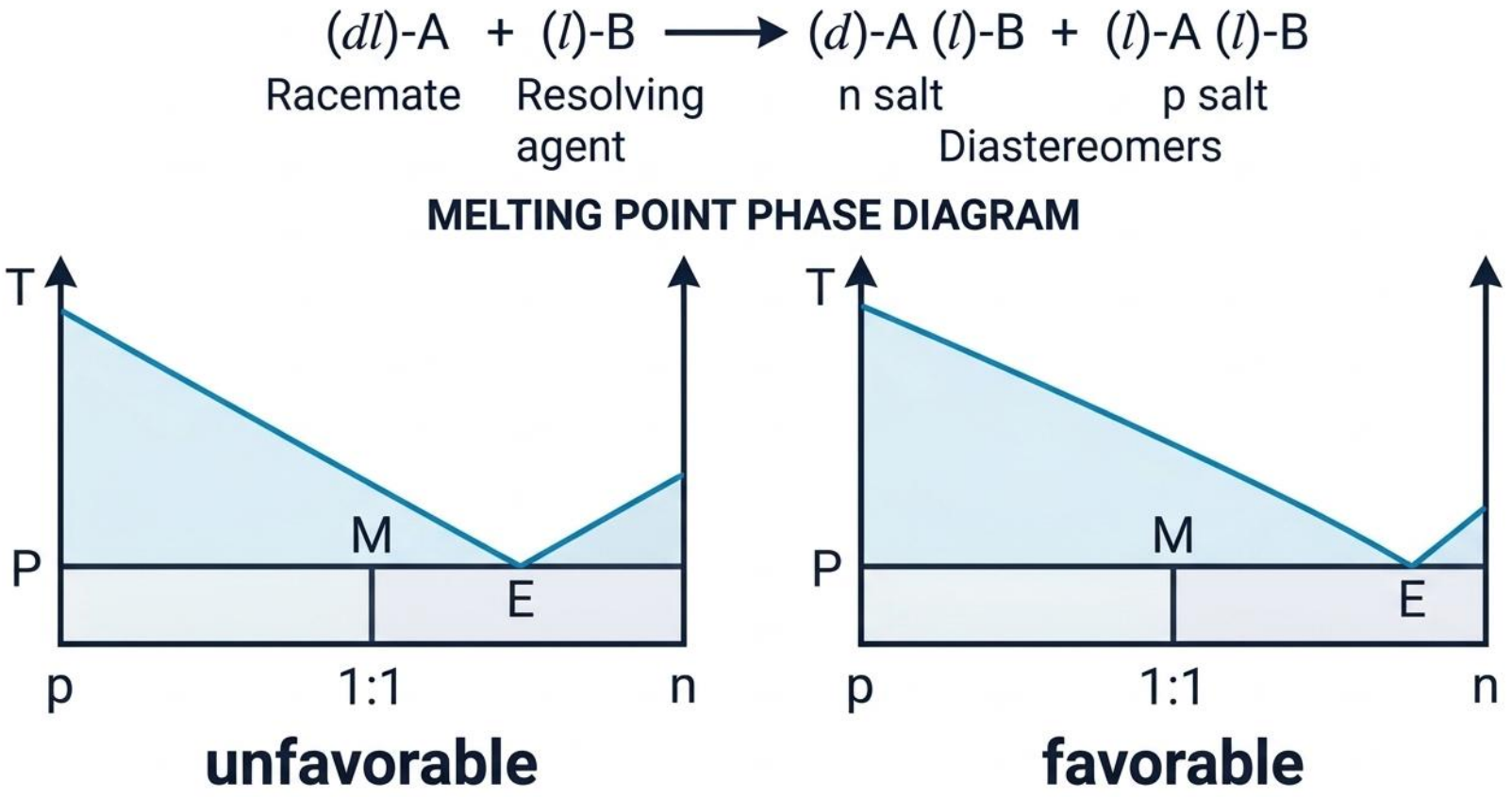


Racemic compound

- Only conglomerates are separable by direct crystallization (lowest melting mixture, eutectic, is the racemate)
- > 90 – 95 % of racemates are racemic compounds

Diastereomer Resolution



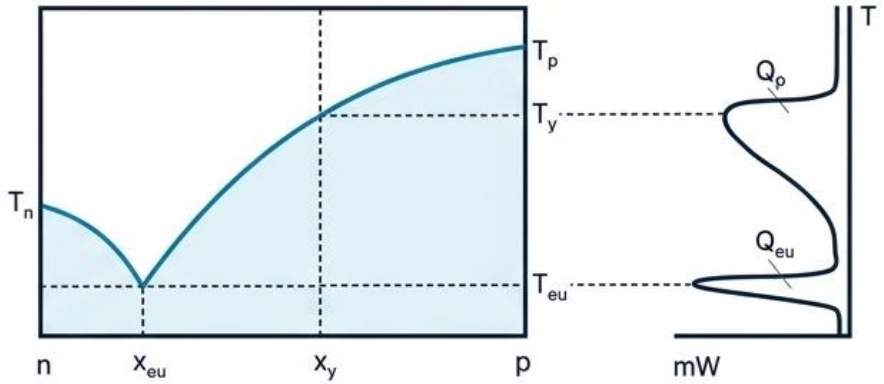


Maximum yield of pure salt = ME/PE

Thermal Determination of x_{eu}



$$\frac{x_y - x_{eu}}{1 - x_{eu}} \ln\left(\frac{x_{eu}}{x_y}\right) = \frac{Q}{R}\left(\frac{1}{T_y} - \frac{1}{T_{eu}}\right)$$



Kozma, D.; Pokol, G.; Ács, M., J. Chem. Soc., Perkin Trans. 2, **1992**, 435.

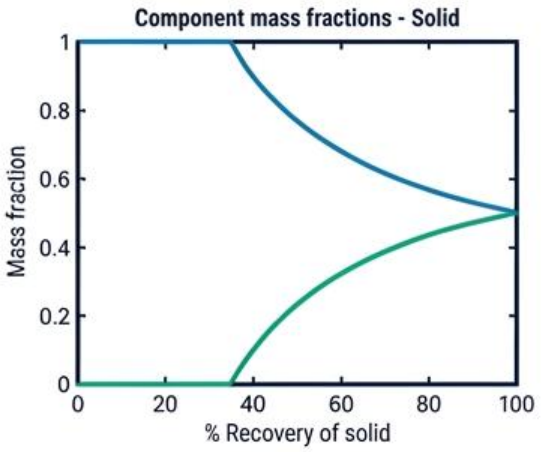
Madarász, J.; Kozma, D.; Pokol, G.; Ács, M.; Fogassy, E., J. Thermal Anal., **1994**, 42, 877.

Ariaans, J.A.; Bruggink, A.; Ebbers, E.; Zwanenburg, B., *Tet. Asym.*,**1998**, 9, 2745.

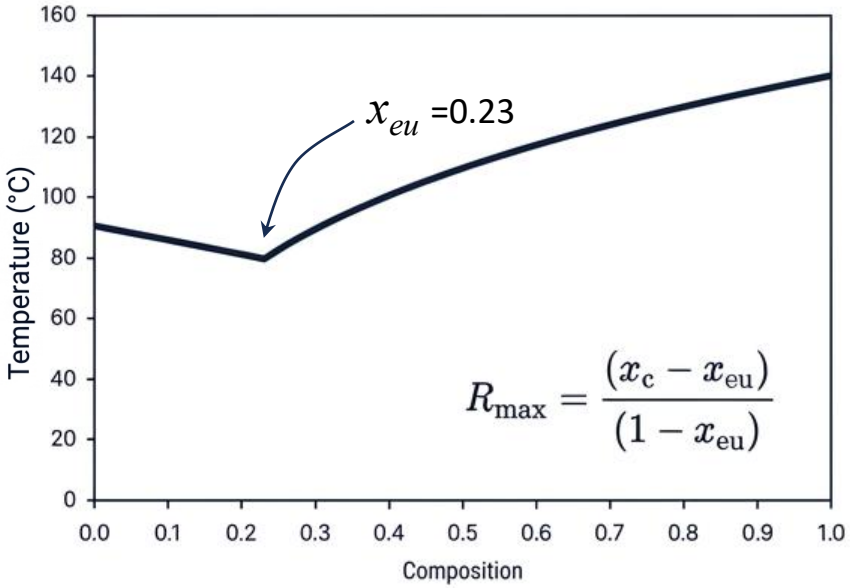
Solution Phase Determination of x_{eu}



- Solution phase measurement
- Dilution can be used to probe for the eutectic

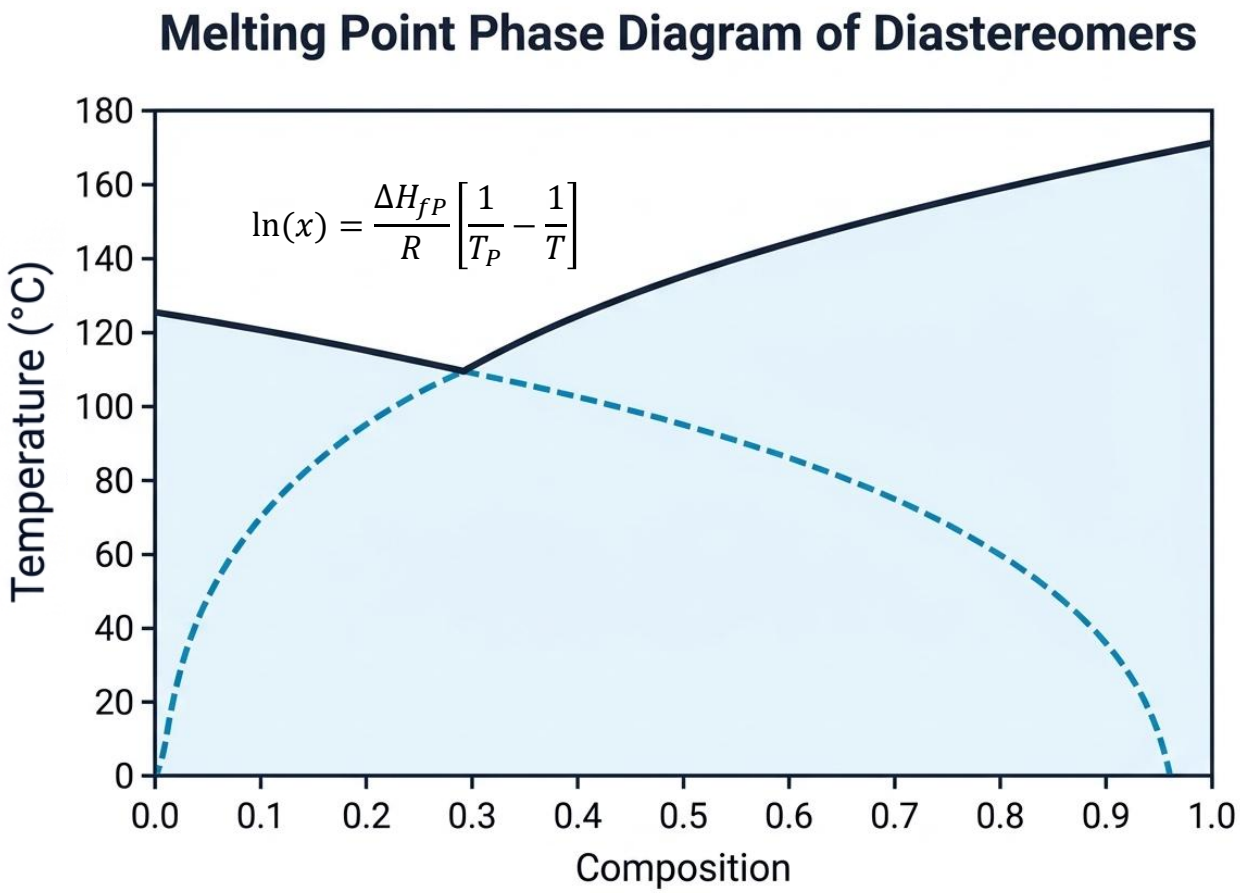


$$R_{\max} = 35\%$$



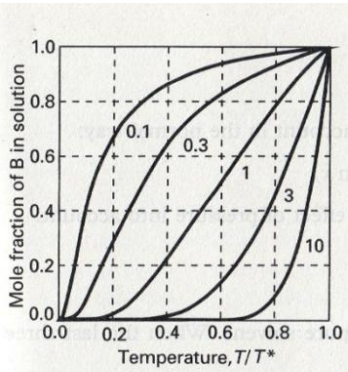
Phase Diagram Puzzles





$$e^{\frac{\Delta H_1}{R}\left[\frac{1}{T_1} - \frac{1}{T_{eu}}\right]} + e^{\frac{\Delta H_2}{R}\left[\frac{1}{T_2} - \frac{1}{T_{eu}}\right]} = 1$$

Physical Chemistry (5th Edition); P.W. Atkins (pp. 226-227)



7.18 The variation of solubility (the mole fraction of solute B in a saturated solution) with temperature (T^* is the freezing temperature of the solute). Individual curves are labelled with the value of $\Delta_{fus}H/RT^*$.

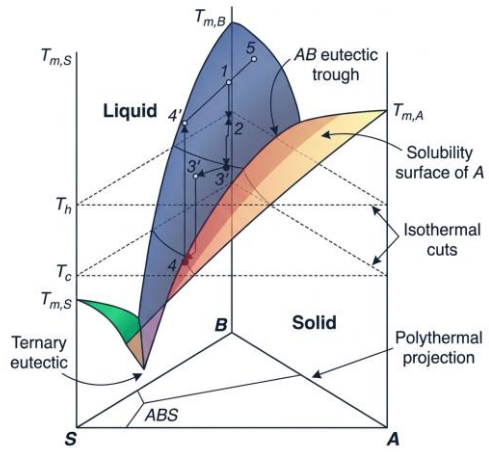
equation to

$$\ln x_B = \frac{\mu_B^*(s) - \mu_B^*(l)}{RT} = -\frac{\Delta_{fus}G}{RT}$$
$$= -\frac{\Delta_{fus}H}{RT} + \frac{\Delta_{fus}S}{R}$$

At the melting point of the solute T^* we know that $\Delta_{fus}G = 0$, and thus $\Delta_{fus}G/RT^* = 0$ too; consequently, the term $\Delta_{fus}G/RT^*$ may be added to the right-hand side. If the enthalpy and entropy of melting are constant over the temperature range of interest, then the entropy terms cancel and, in much the same way as before, we obtain

$$\ln x_B = -\frac{\Delta_{fus}H}{R}\left(\frac{1}{T} - \frac{1}{T^*}\right) \qquad (20)^\circ$$

This expression (which is plotted in Fig. 7.18) shows that the solubility of B decreases exponentially as the temperature is lowered from its melting point and that solutes with high melting points and large enthalpies of melting have low solubilities at normal temperatures.



Multicomponent Eutectics



Ternary

$$e^{\frac{\Delta H_1}{R}\left[\frac{1}{T_1}-\frac{1}{T_a}\right]}+e^{\frac{\Delta H_2}{R}\left[\frac{1}{T_2}-\frac{1}{T_a}\right]}+e^{\frac{\Delta H_3}{R}\left[\frac{1}{T_3}-\frac{1}{T_a}\right]}=1$$

Binary

$$e^{\frac{\Delta H_1}{R}\left[\frac{1}{T_1}-\frac{1}{T_b}\right]}+e^{\frac{\Delta H_2}{R}\left[\frac{1}{T_2}-\frac{1}{T_b}\right]}=1$$

Margerum, J. D.; Lackner, A. M., Mol. Cryst. Liq. Cryst., 1981, 76, 211.

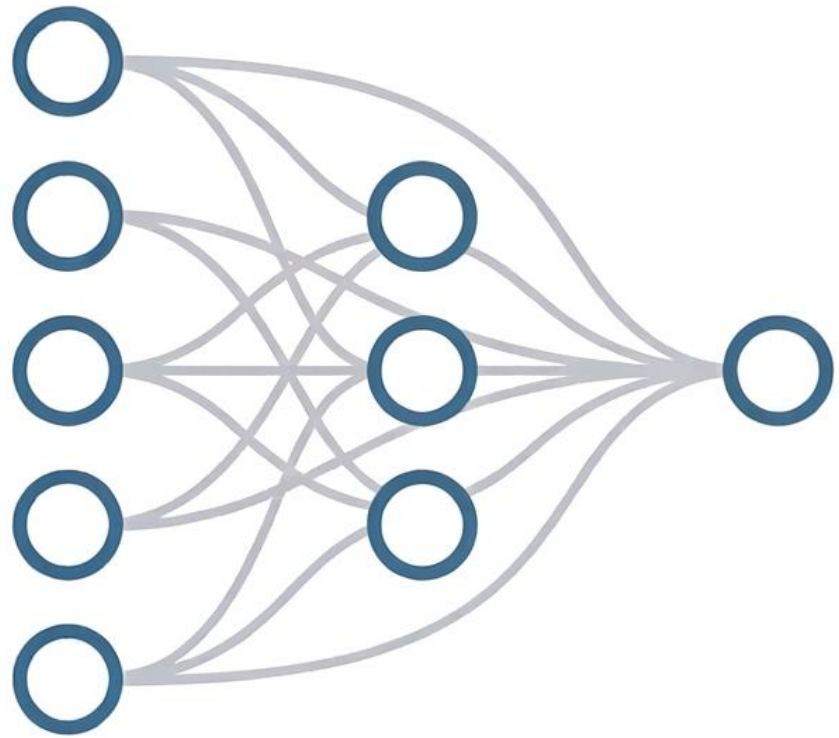
Illustrated MultXeu Calculation



R_{max} = 44.01%

Recoveries				Component 1	Component 2	Component 3	Component 4	Component 5
			MPt (Ti) / °C	131.0	133.6	179.9	131.70	126.0
			Heat of Fusion (ΔHi) / kJ/mol	35.38	24.95	31.19	21.28	26.23
			Starting Composition	69.7%	8.7%	10.1%	4.5%	6.9%
Stage 1	85.03%	**A**	Eutectic MPt: 67.4 °C, Composition	13.94%	23.76%	6.46%	30.26%	25.59%
		B	Composition After Eutectic Removal	79.55%	6.08%	10.79%	—	3.58%
Stage 2	89.98%	**A**	Eutectic MPt: 80.1 °C, Composition	21.91%	32.69%	9.63%	—	35.78%
		B	Composition After Eutectic Removal	85.96%	3.11%	10.92%	—	—
Stage 3	93.44%	**A**	Eutectic MPt: 96.3 °C, Composition	37.19%	47.47%	15.34%	—	—
		B	Composition After Eutectic Removal	89.39%	—	10.61%	—	—
Stage 4	61.55%	**A**	Eutectic MPt: 119 °C, Composition	72.39%	—	27.61%	—	—
		B	Composition After Eutectic Removal	100.00%	—	—	—	—

Smith A. A., *Tet. Asym.*, **1998**, 9, 2925 [see therein for a summary of MultXeu algorithm]



NODES

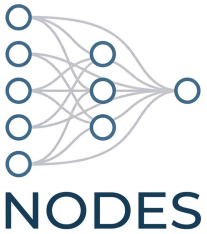
Python source code: https://github.com/idiocratease-Iggy/nodes

Scope of NODES



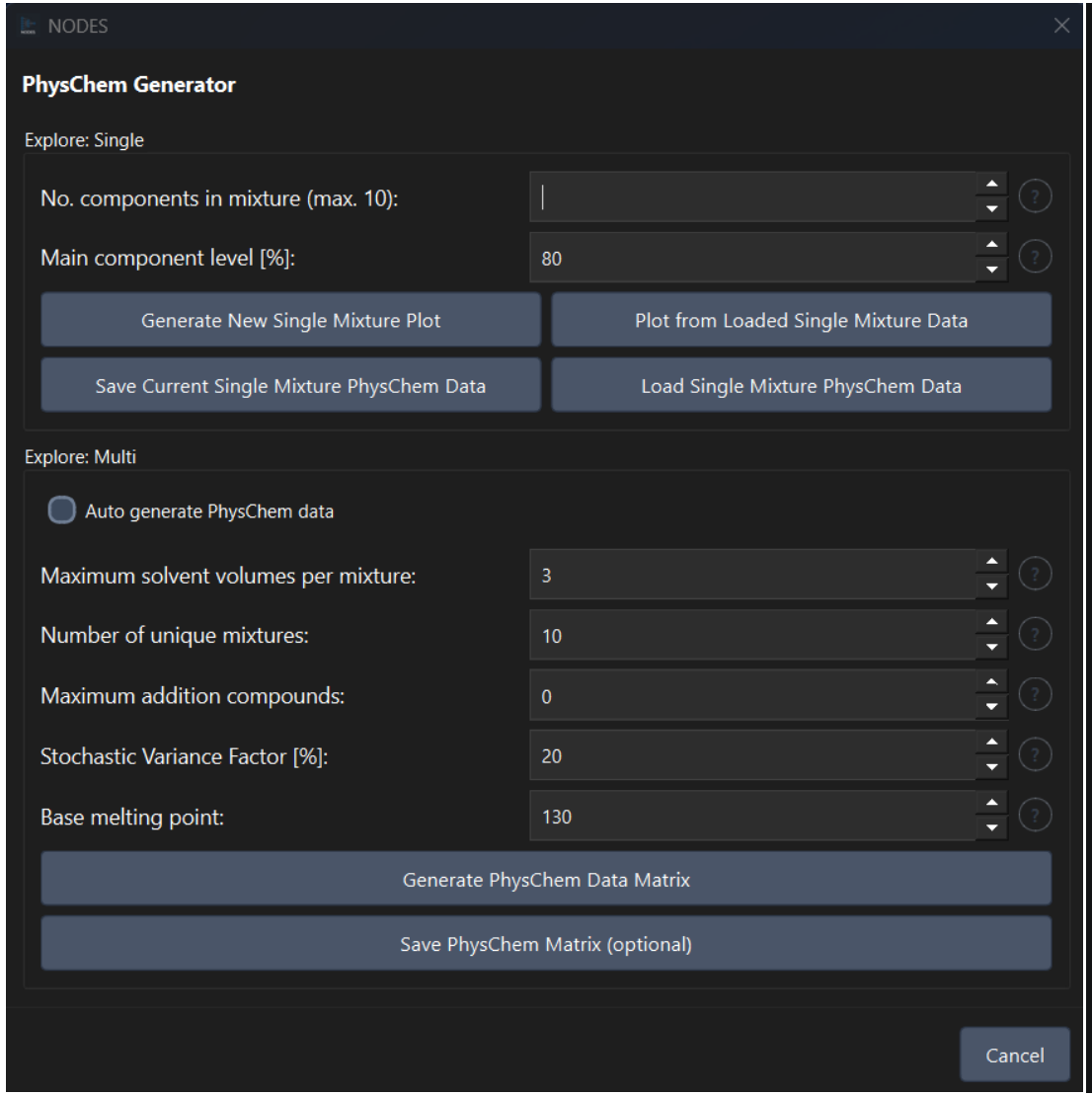
In Silico data sets with known R_{max}

- Explore: Single
  - Automatically generate PhysChem data for up to 10 component mixtures
  - Plots the predicted purification profile
  - Save & read single mixture PhysChem data JSON files



- Explore: Multi
  - Create data sets for up to 10 components
  - Simulates multi solvent-volumes scenarios
  - Up to 99 unique starting mixtures
  - Simulates random analytical & systematic cross-contamination error
  - Creates multiple excel files: each solvent volume set and with-without error
  - Select well-plate or free-format write, plus replicates & plate-filling logic
  - Standardized file naming (YYMMDD_MMSS_Vol0N_errXpc.xlsx)
  - Save & read PhysChem JSON files

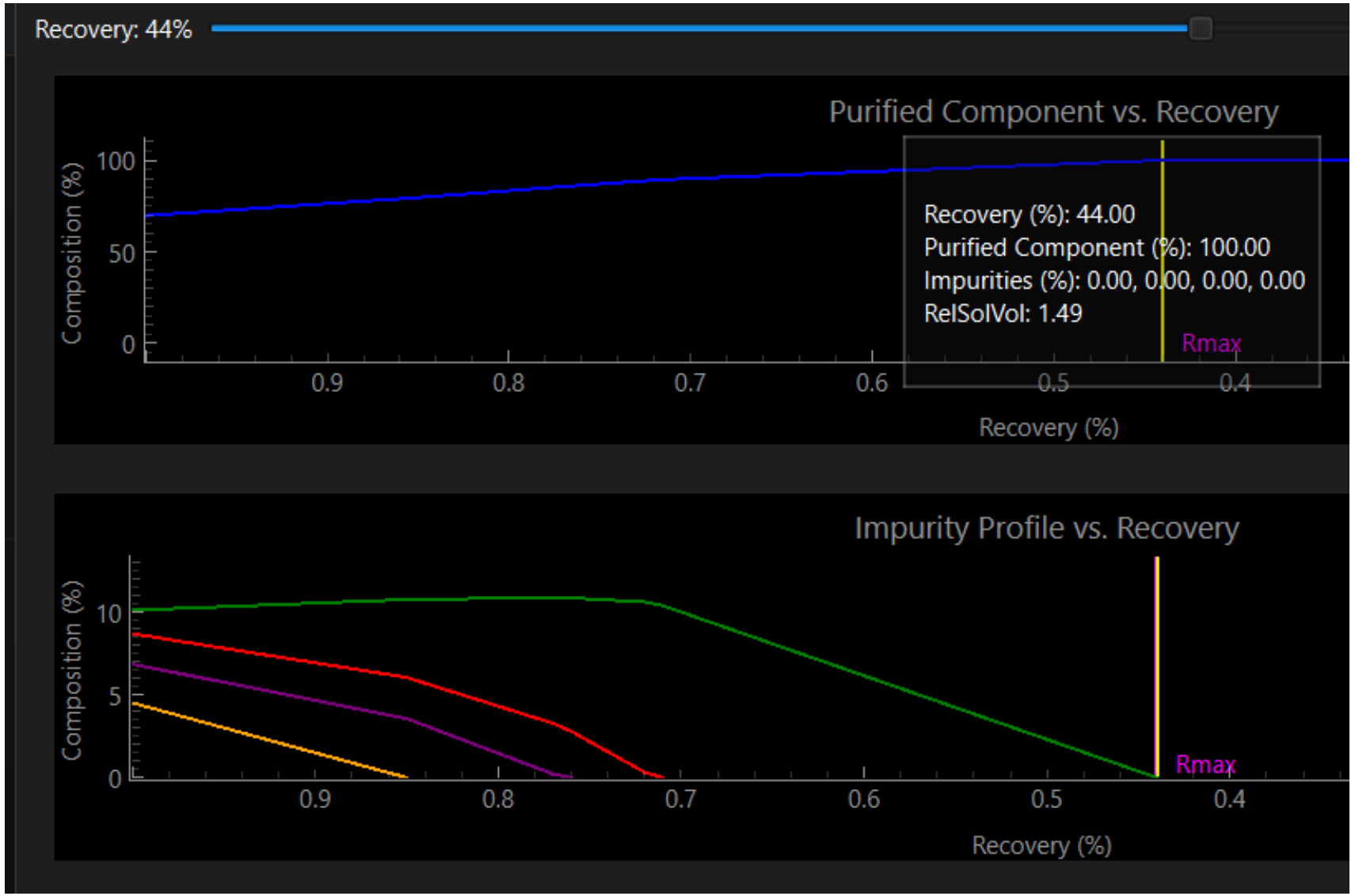
NODES App



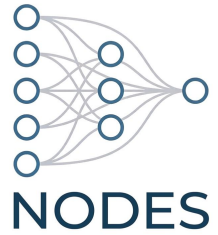

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NODES

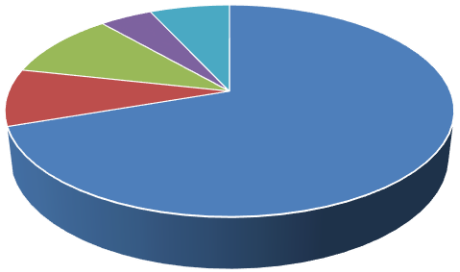
NODES Stacked Plot



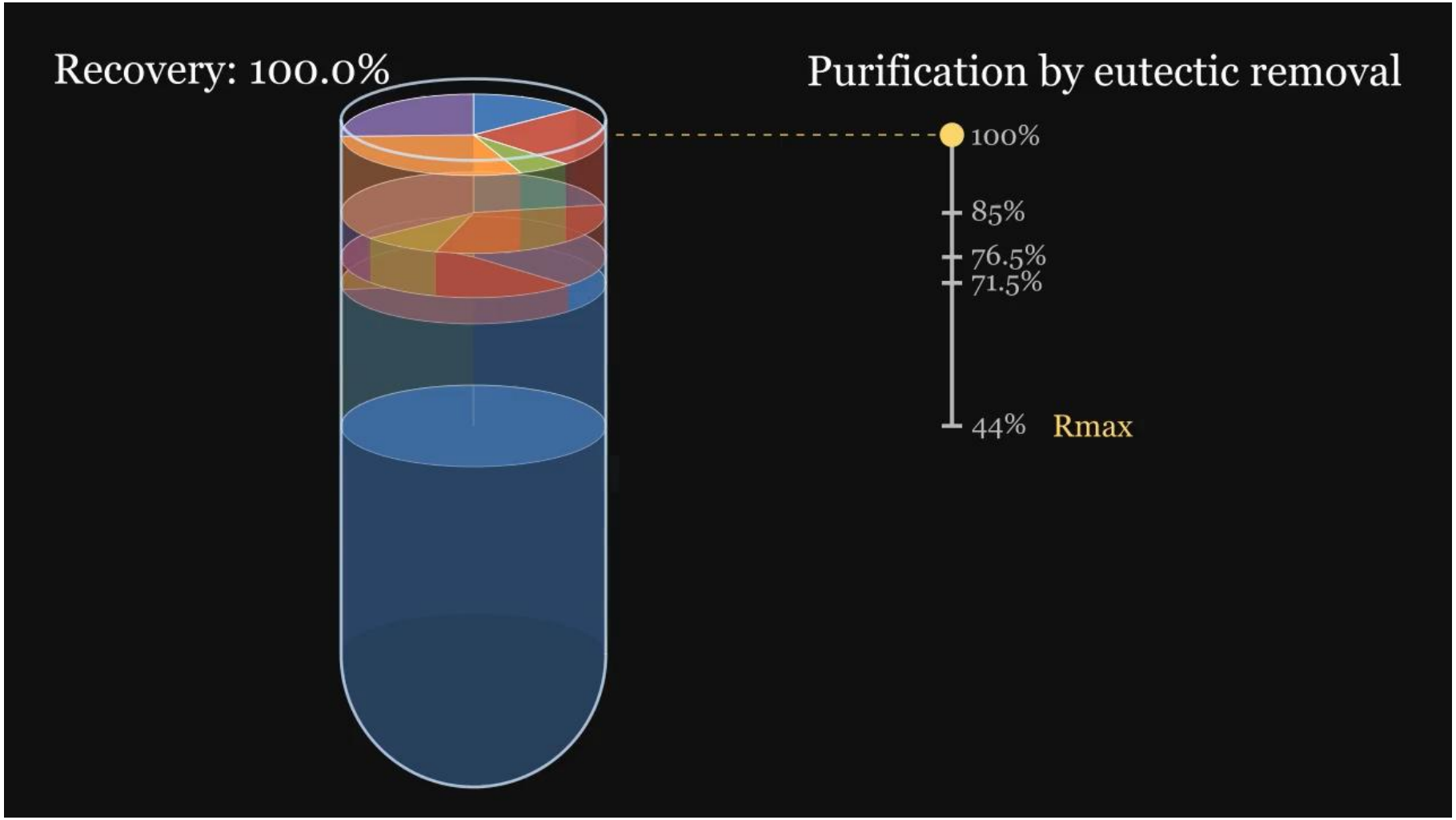




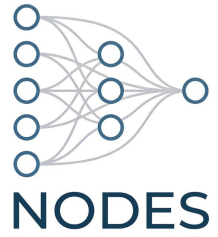
Starting Composition (Xc)



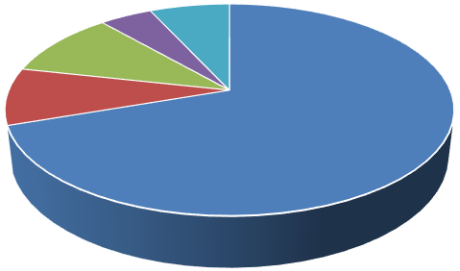
Focussing on the Eutectics







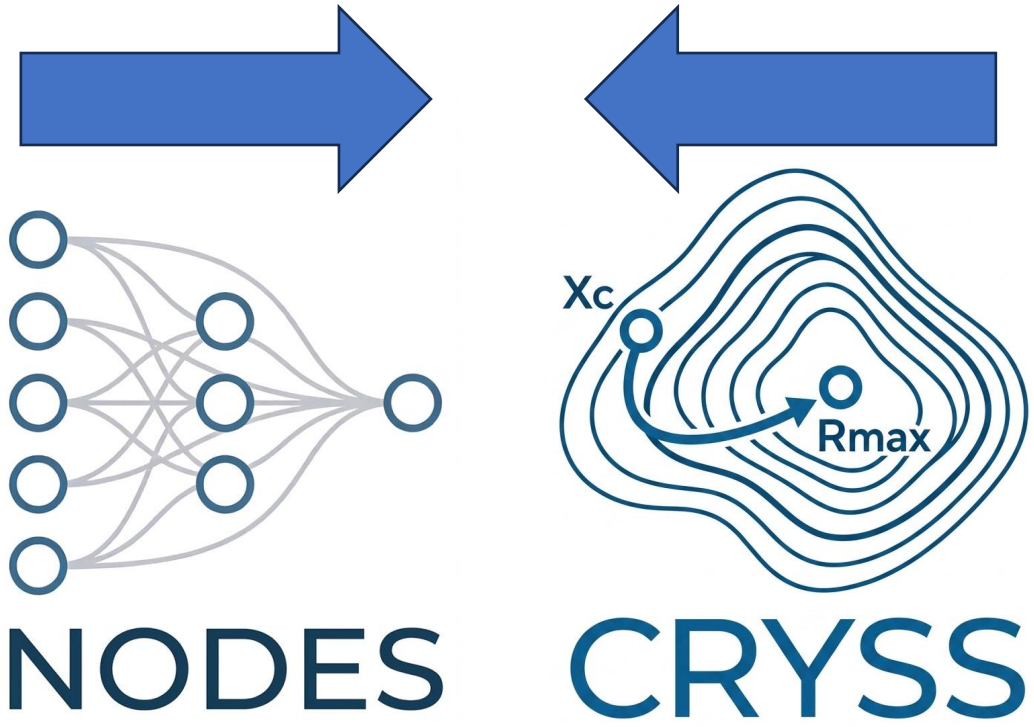
Starting Composition (Xc)



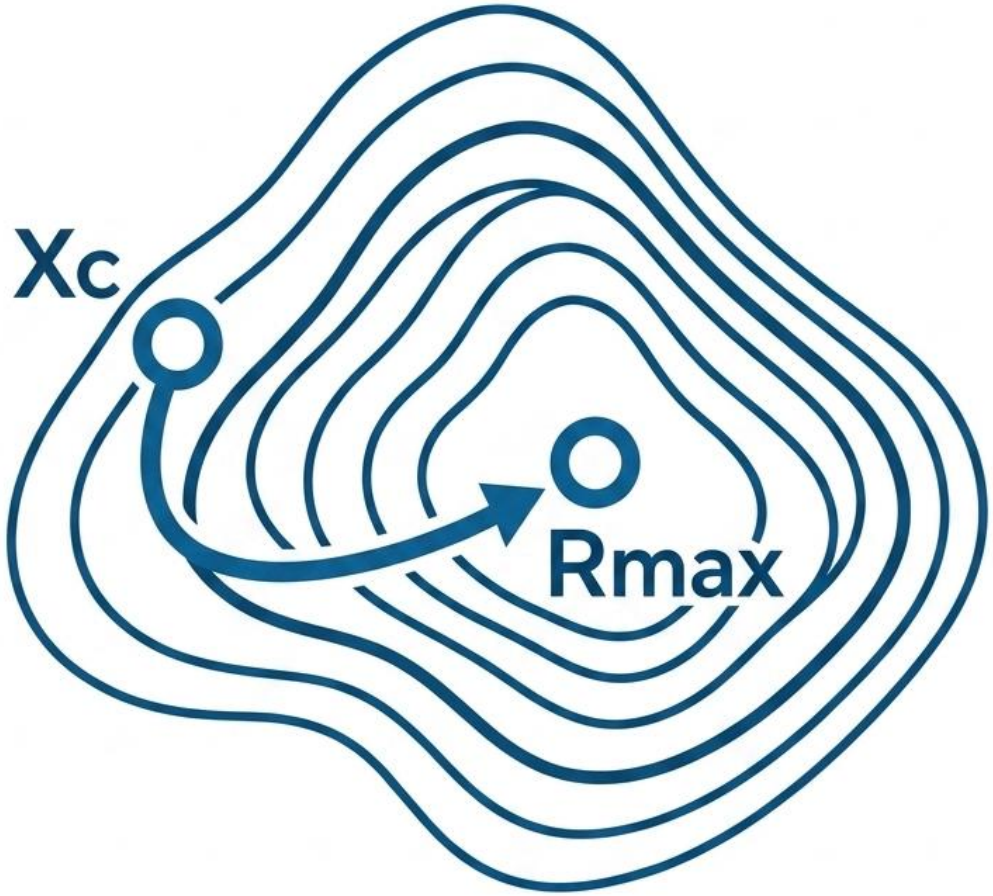
https://www.manim.community/

NODES - Precursor to CRYSS





- NODES elaborates the 1998 MultXeu VBA algorithm in Python
- NODES uses PhysChem data to create *in silico* partitioned data-set deterministic R_{max} calculation
- CRYSS implements MultXeu core algorithm
- CRYSS deconvolutes equilibrium partitioned data, then predicts R_{max}

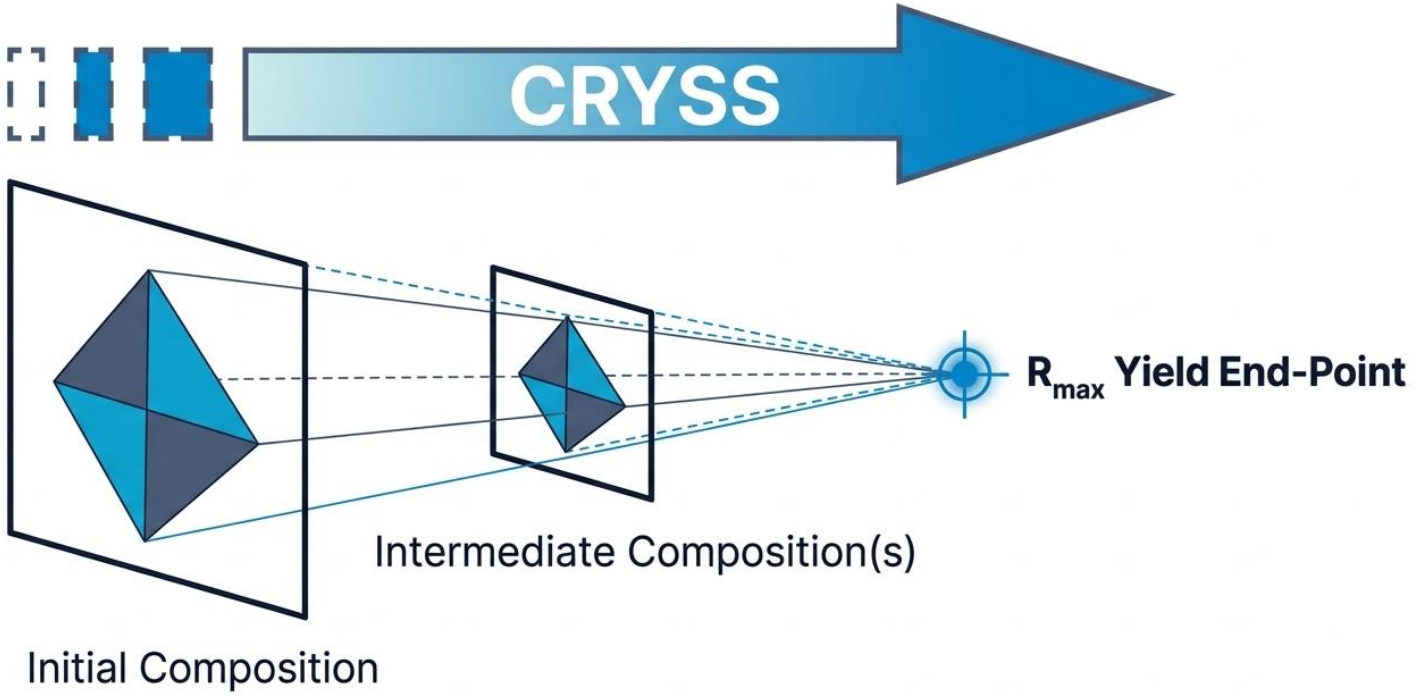


CRYSS

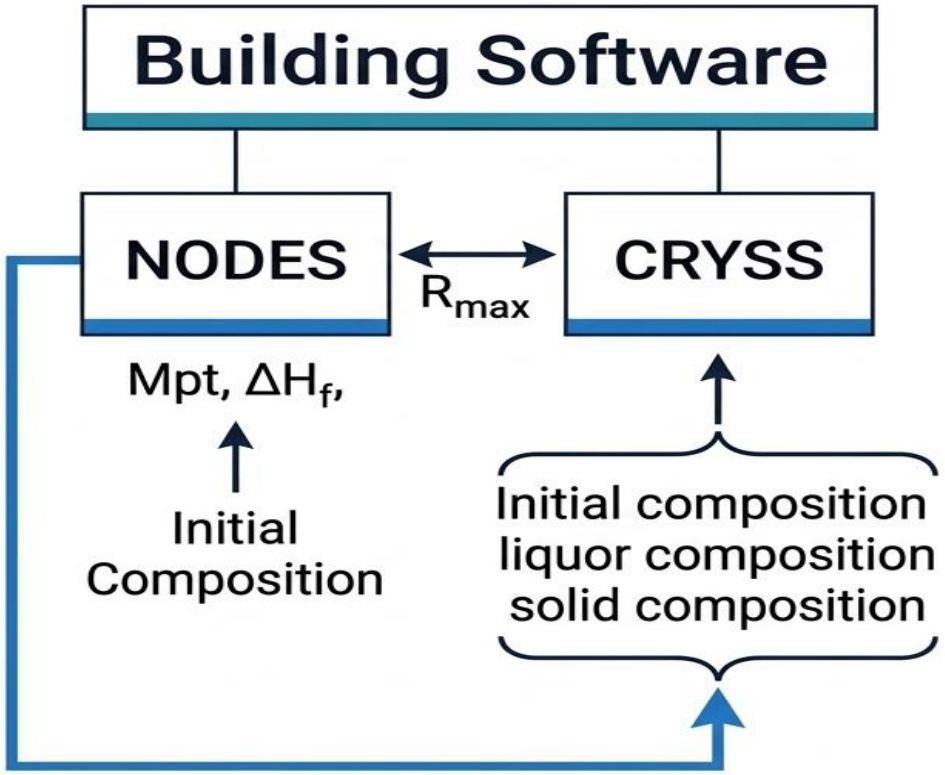
Python source code: https://github.com/idiocratease-Iggy/CRYSS-Release

The CRYSS Approach

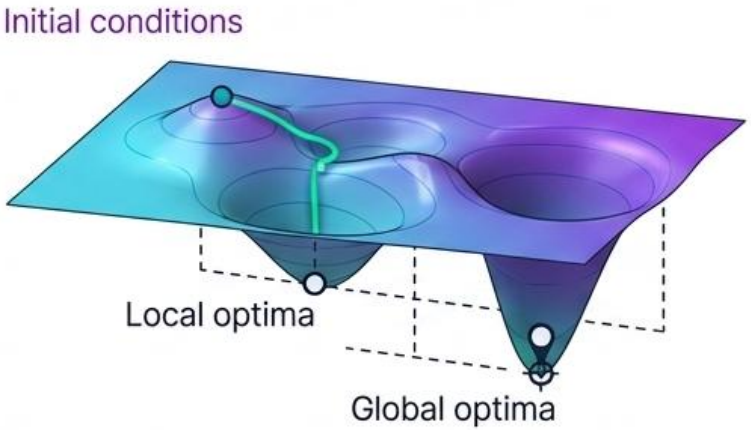
- Mechanistic model & fitted to experimental data
- Only requires data from partial purifications
  - Rank systems
  - Predict solubility diagram
  - Optimize process in minimum steps



CRYSS: MultXeu + Optimizer

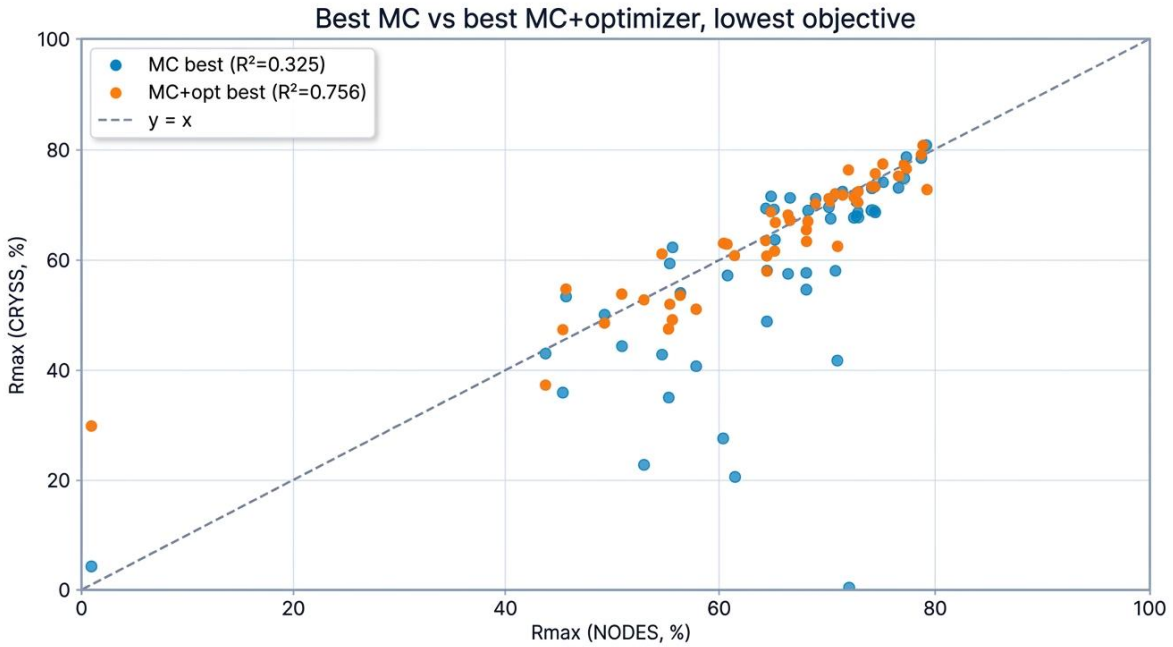


Beware of issues with local minima



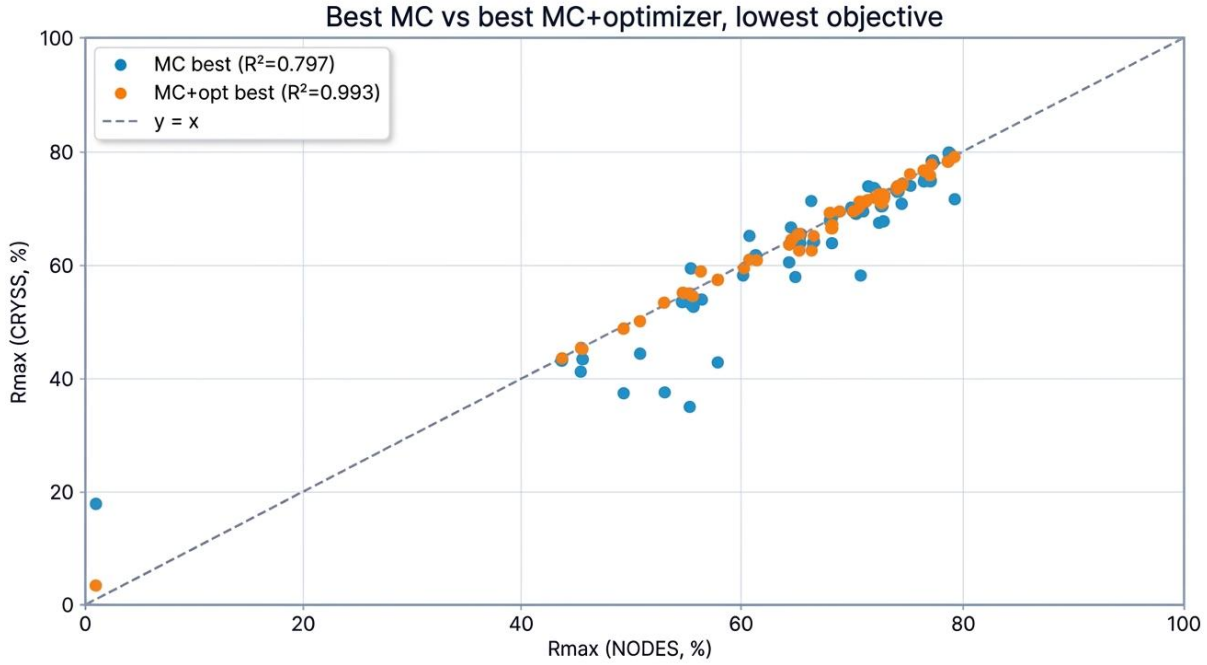
1) Alan A. Smith, Ben McKay, Eric W. P. Damen, Simone Darphorn-Hooijschuur, Erik-Jan Ras, Goran Verspui; Prediction of Maximum Yield in the Crystallization of Multicomponent Isomeric Systems. Org. Process Res. Dev. 17 November 2006; 10 (6): 1132–1143.
2) US 7,025,828 B2, 11-Apr-2006

Results: Single vs. Three Solent Volumes

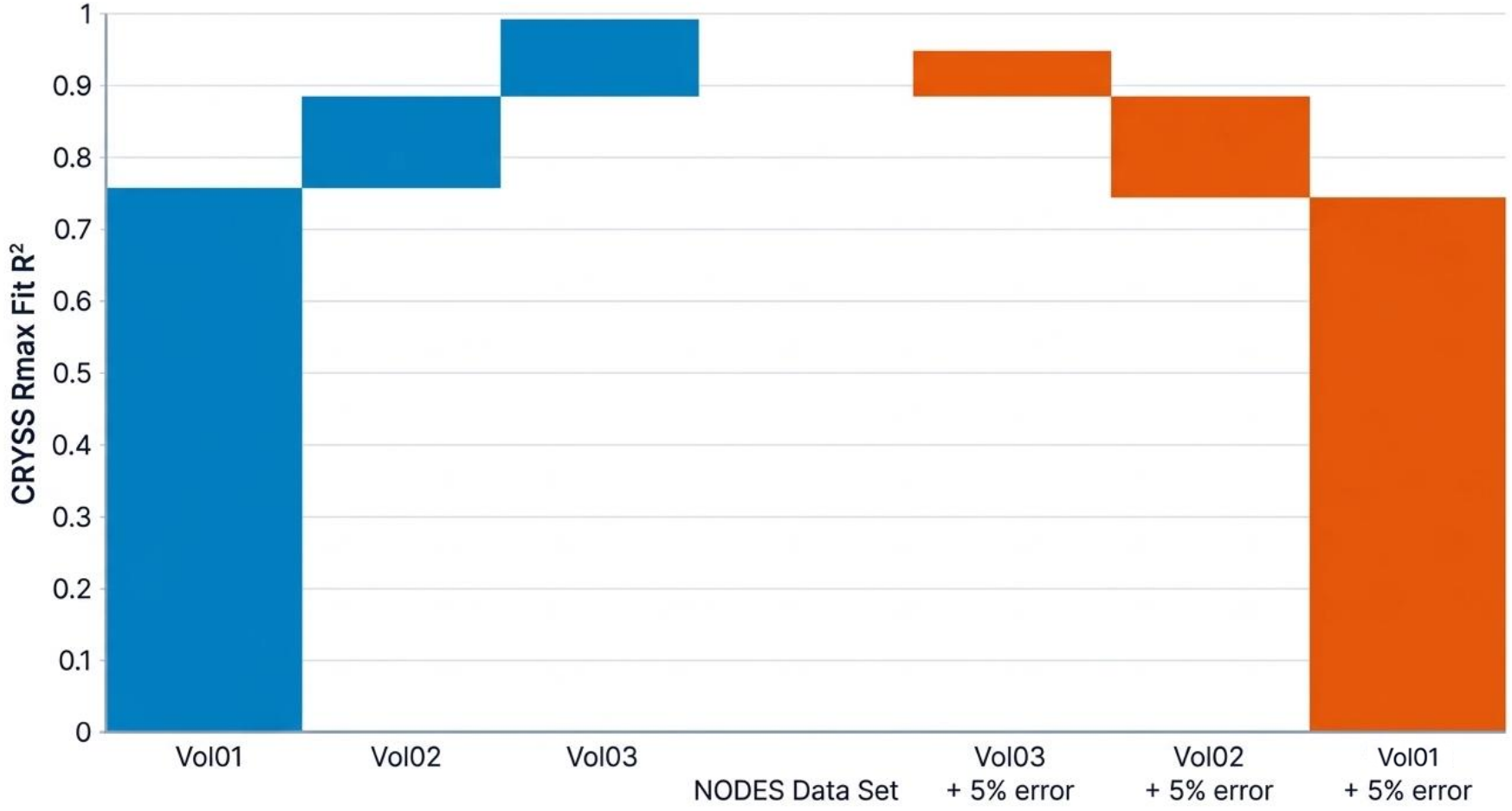


Single solvent volume composition slice

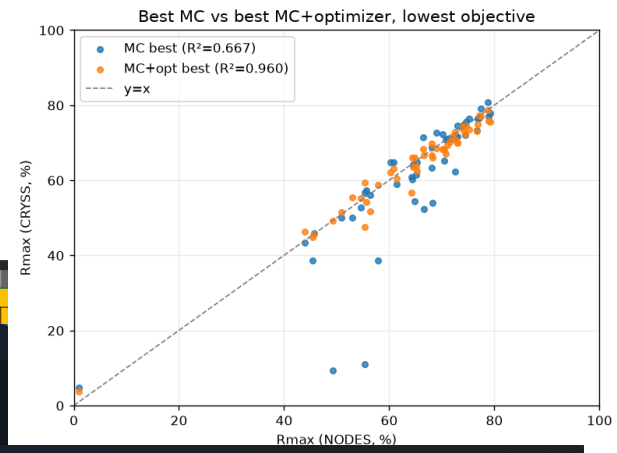
Three solvent volume composition slice

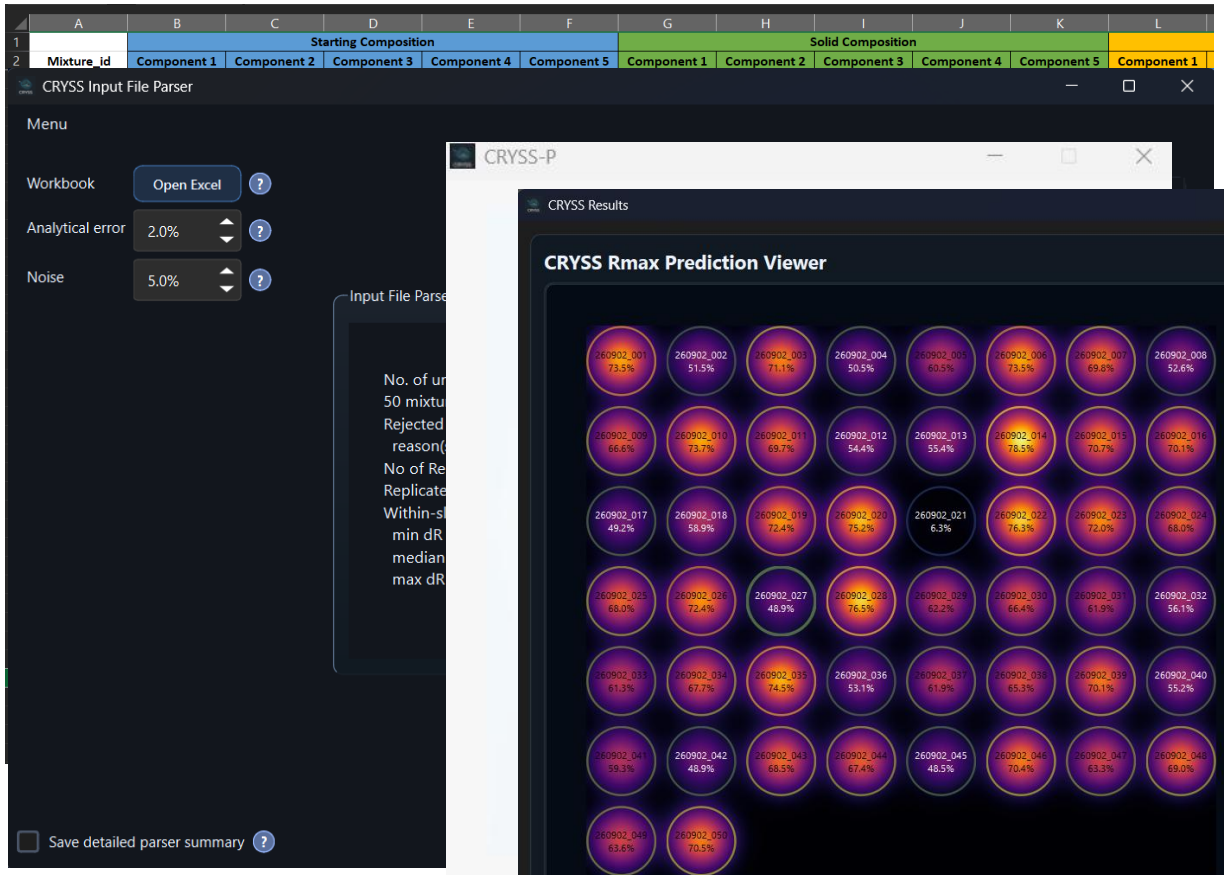


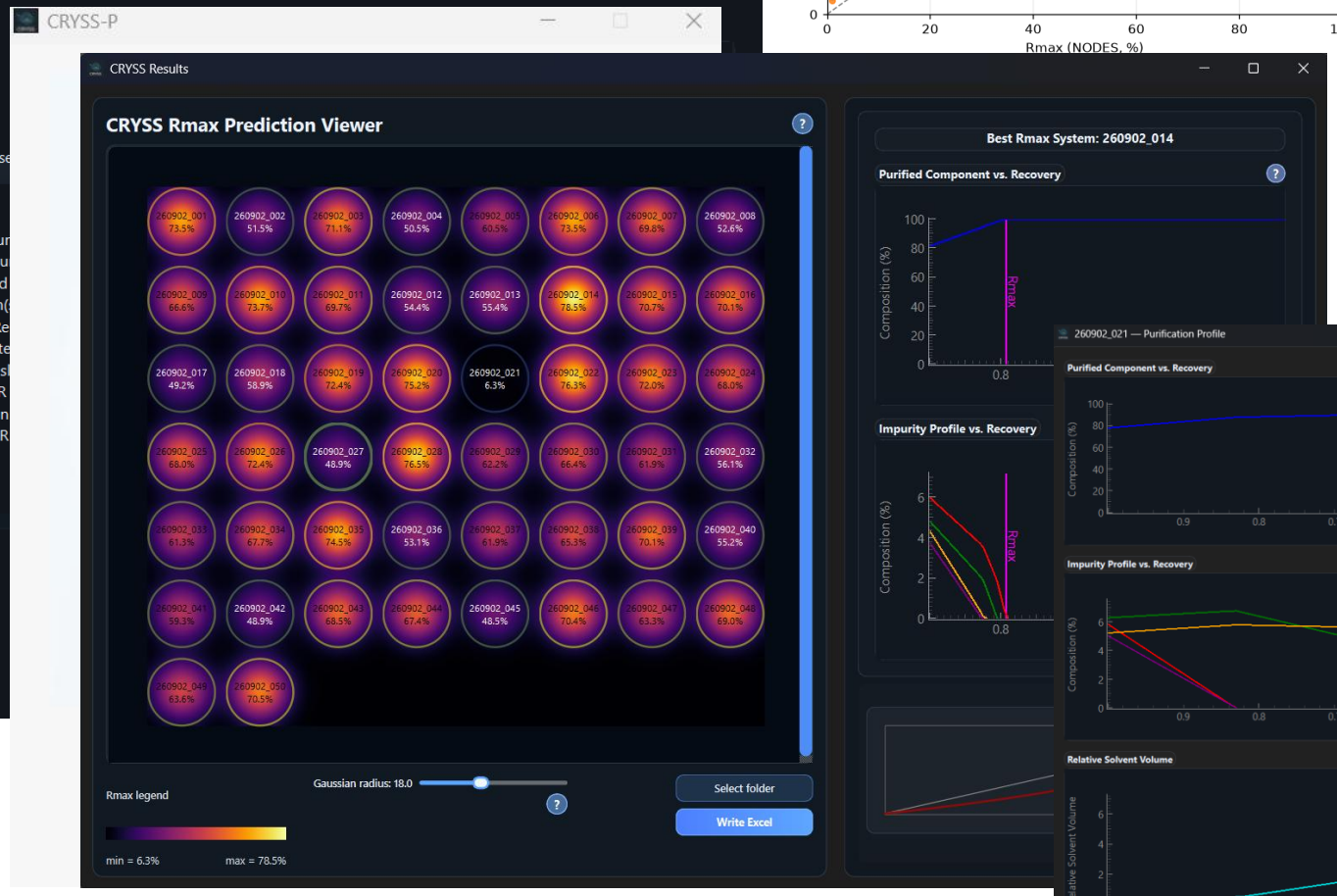
Results: R^2 vs Solvent Vols & Error



CRYSS Workflow









https://github.com/idiocratease-Iggy/CRYSS-Release/releases/tag/V2.0

RELEASE V2.0: MIT License - it is free for academic, industrial, and commercial use.

Applicability to Real-World Systems



Use of activity to capture deviation from ideality

$$a_i = \gamma_i \cdot x_i = e^{Q} \quad ; \quad \text{where } Q = \frac{\Delta H_i}{R} \cdot \left(\frac{1}{T_i} - \frac{1}{T}\right)$$

Since the mass fraction is related to mole fraction:

$$m_i = x_i \cdot M_i \cdot \sum_j \frac{m_j}{M_j} \quad \text{Or written more simply:} \quad m_i = \beta_i \cdot x_i$$

Since: $m_i = \beta_i \cdot x_i$, you end up with $a_i = \gamma_i \cdot x_i = \gamma_i \cdot \frac{m_i}{\beta_i} = e^{Q}$

$$\ln(m_i) = Q + \ln(\beta_i) - \ln(\gamma_i)$$

Or:

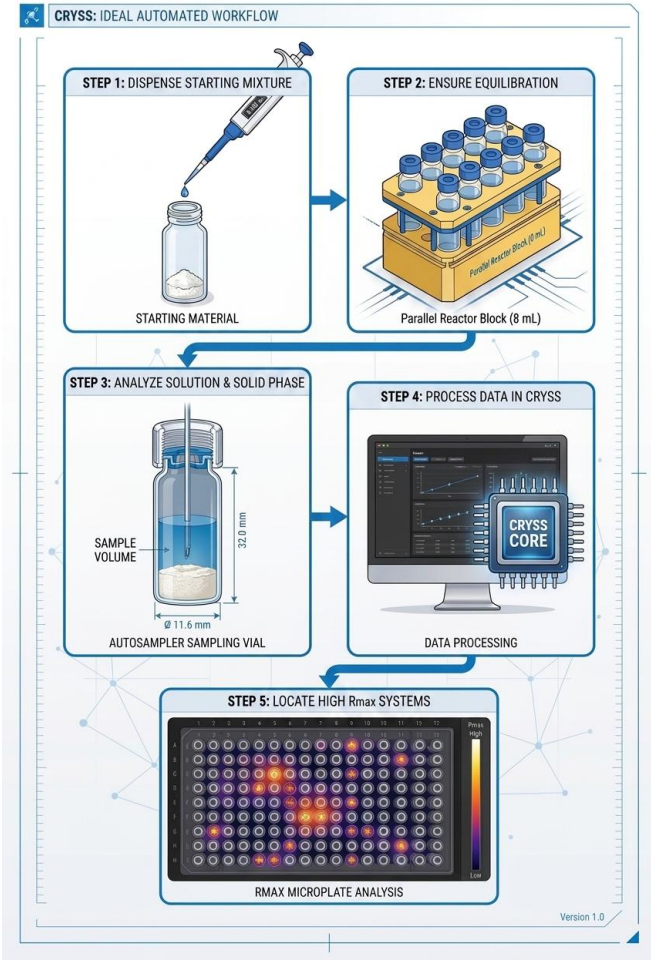
$$m_i = e^{Q + \ln(\beta_i) - \ln(\gamma_i)}$$

PARAMETER DEFINITIONS

- a_i: activity of solute in solution
- γ_i: activity coefficient of solute in solvent
- x_i: mole fraction of solute: $x_i = \frac{m_i}{M_i} / \sum_j \frac{m_j}{M_j}$, M_i molar mass
- ΔH_i: enthalpy of fusion of pure solid solute
- T_i: melting temperature (K) of pure solid solute
- R: ideal gas constant
- T: absolute temperature (K) of system

Since in CRYSS any deviations from ideality are easily accommodated, and avoids the need to supply data in mole fraction, mass/composition data works fine.

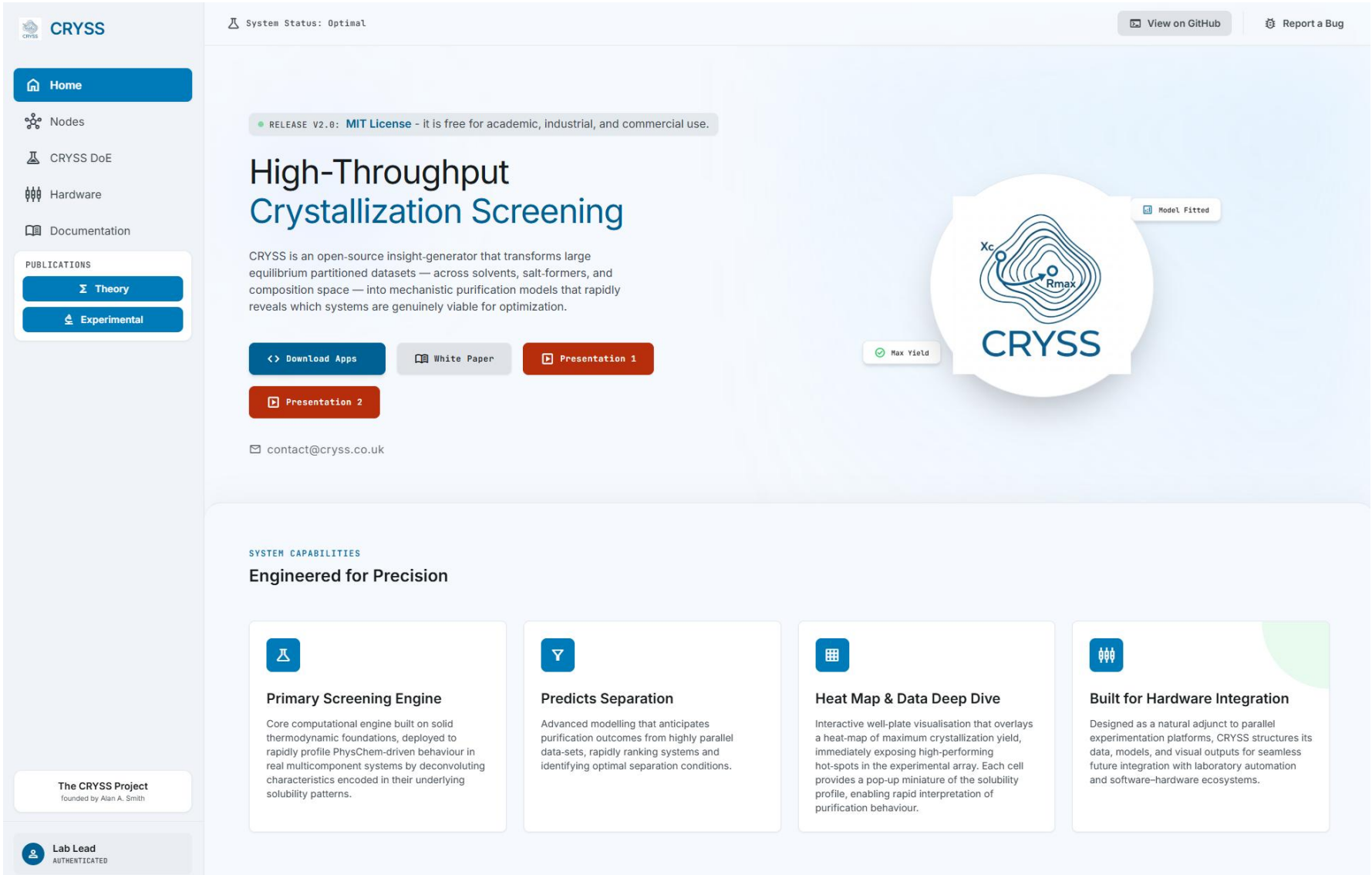
Practical Considerations





- Small Scale
  - Liquor sampling & w/w method → Recovery (R)
  - Avoid solid contamination
  - Lever-rule to x_c, x_l /R → x_s
- Larger Scale
  - Slurry sampling, filtration, x_s , x_l analysis possible
  - R calculated *via* lever rule
- Ensure equilibrium
- Avoid solvent loss
- Accuracy is key and well suited to automation
- Include replicates (parsed to upgrade input data)

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Acknowledgements



- Eric Damen
- Simone Darphorn-Hooijschuur
- Francois Gilardoni
- Ben McKay
- Lisa McQueen (née Agocs)
- Erik-Jan Ras
- Goran Verspui